

Sat_AI Forest Watch

Scalable Deep Learning for Real-Time Forest Monitoring in India

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Deforestation remains a critical environmental challenge in India, with severe impacts on biodiversity, carbon sequestration, and sustainable development. Timely and accurate detection of forest cover loss is therefore essential for effective conservation planning and policy interventions. This study presents SatAI-Forest Watch, a scalable deep learning framework for near real-time deforestation detection using multi-temporal Sentinel-2 imagery. Sentinel-2 offers high spatial resolution (10 m) and rich spectral information across 13 bands, enabling robust monitoring of forest cover dynamics. Multi-seasonal composites were generated through Google Earth Engine(GEE) to capture temporal variations in vegetation, while deep learning architectures were employed for semantic segmentation and change detection across diverse forest regions in India. Vegetation indices such as NDVI and NBR were integrated with multi-temporal stacks to enhance sensitivity to subtle disturbances.

Unlawful forest clearance erodes biodiversity, accelerates climate change, and undermines local livelihoods—yet it often goes undetected until the damage is irreversible. Traditional monitoring methods rely on manual reporting, sporadic patrols, or costly proprietary platforms, which makes near-real-time tracking difficult, especially across vast and remote regions. This project tackles that gap by building an AI-driven pipeline that detects forest cover loss from satellite imagery and turns it into actionable alerts for rapid response.

To achieve this, we will leverage open-access satellite datasets such as: Sentinel-2, Global Forest Change Dataset.

Title	Demerits
Remote Sensing-Based Analysis of Deforestation Trends in Biodiversity Hotspots	The study lacks ground verification, socio-economic analysis, and predictive modelling, relying heavily on satellite data
Monitoring Deforestation in the Eastern Himalayas Using Satellite Data and Deep Learning	Data accessibility challenges.
Predicting Deforestation Risk Using Machine Learning and Satellite Imagery	Depends on historical data, which may not reflect sudden changes.

Table 1: Tools Watching Over Our Forests

- Google Earth Engine for processing
- NDVI and NBR vegetation indices
- U-Net or DeepLab for semantic segmentation
- Python with TensorFlow or PyTorch
- Rasterio / GDAL for geospatial data handling

Packages

- Keras, torchvision
- scikit-image
- geopandas
- EarthPy

Libraries

- TensorFlow / PyTorch
- OpenCV
- NumPy, Pandas
- Rasterio, GDAL

Smart Eyes on Deforestation

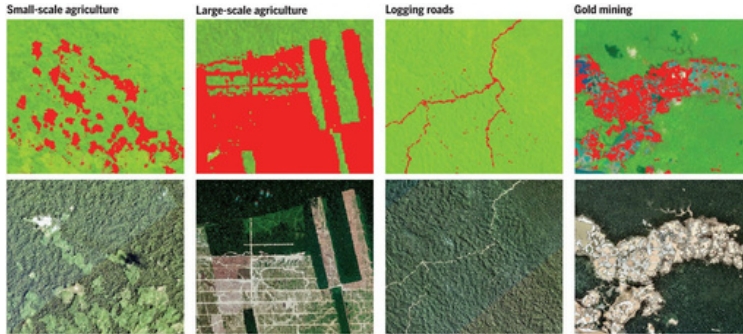


Figure 1: Smart Surveillance for Forest Protection : The output is a pixel-level change map highlighting areas of suspected deforestation. This map is post-processed to reduce noise and grouped into geospatial events. The final stage involves an alert system that sends notifications to NGOs or government agencies, and a dashboard that visualizes the results with before/after imagery, confidence scores, and area summaries.

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